

Finally, it includes a call to the `ended()` method in every frame to see if the time has run out. If the timer ends before the user has solved the puzzle, it plays a sound and displays a message telling the user that they ran out of time.

Summary

This chapter gave you a number of ways to handle timing in programs. I discussed three different approaches: first by counting frames, second by creating a custom event, and finally by remembering a start time and subtracting it from the current time to get the time elapsed.

Using the third approach, we built a generic reusable `Timer` class (which you can find in the `pyghelpers` package). I also showed two additional classes from this package, `CountUpTimer` and `CountDownTimer`, that can be used to handle timing in programs where you want to show a timer to the user.